

Four-Element Same-Order Residual Obstruction

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Pass 114 tests the Pass-113 MacNeille witness for a same-carrier, same-order residuated tensor expansion. The witness has carrier $\{0, a, b, c\}$ and order

$$0 < a < b, \quad 0 < c,$$

with b incomparable to c . It satisfies finite A1-A4, G2, and FG2 as a table check, has no syntactic `boxtimes` fixed point, and has the non-principal MacNeille fixed cut $\{0, a, b, c\}$.

The residual search keeps this exact carrier and order fixed. For every possible unit, it enumerates every binary operation with that unit as a two-sided unit, then keeps operations that are associative, monotone in both arguments, and admit both residuals by the principal-downset criterion.

The exhaustive counts are:

unit	operation space	associative	associative+monotone	full residual
0	262144	156	0	0
a	262144	156	6	0
b	262144	156	18	0
c	262144	156	32	0

Across all units, 624 tensors are associative and 56 are associative and monotone, but none admits both residuals. The first named obstruction among the surviving units is the fiber

$$\{x : 0 \otimes x \leq 0\} = \{0, a, b, c\}.$$

This is the whole carrier. Since the carrier has no greatest element, the fiber is not a principal downset and cannot be represented by a residual element.

The result is therefore a fixed-carrier and fixed-order no-go: the Pass-113 witness is not yet a residuated APS witness. It does not rule out order repairs, completion-level tensors, or other four-element witnesses. The next test is to add the weakest top/join repair that makes the whole-carrier fiber principal, then re-check finite A1-A4, G2, FG2, and MacNeille separation.

Machine artifacts: `code/scripts/check-pass114.py`, `artifacts/reports/pass114-four-element-residual-boundary-check.json`, and `artifacts/reports/pass114-four-element-witness-residuated-tensor-search.json`.